

THERMOPAC · ENGINEERING REVIEW · DESIGN 269

Kühni turbine–stator successor geometry

Read-only Stage-3 audit and proposed double-entry shrouded turbine /
perforated-stator development

**UNAPPROVED SUCCESSOR GEOMETRY
PRELIMINARY — NOT FOR FABRICATION**

Stage-3 run **64** · Stage-4 calculation **13**

Report issue: engineering review candidate only; no new rule-set version assigned.

Audit PASS is a software-selection verdict.

The saved operating point is **SCALE_UP_EXTRAPOLATION**. Neither the audit nor geometric area closure validates the successor hydrodynamics.

Confirmed saved basis: Ø600 mm column, Ø198 mm turbine, 180 mm pitch, empirical $\phi = 0.40$, **45 RPM**, NMP continuous / RRBO dispersed.

This report preserves the existing R1 implementation and immutable revisions. Candidate dimensions are explicit engineering choices for review—not released CAD or fabrication instructions.

All dimensions: mm; areas: mm², unless stated otherwise. Effective areas are geometric, not discharge-coefficient-adjusted hydraulic areas.

Report sequence and decision boundary

1. Stage-3 selection audit and applicability limits
2. Confirmed inherited diameter and saved basis
3. Complete turbine topology and construction evidence
4. Turbine eye candidate and sensitivity
5. Upper/lower entry sections and obstruction accounting
6. Stator centre-opening proposal
7. Stator gross/net area and perforation development
8. Clearance, ligament and assembly checks
9. Proposed dimensioned review drawings
10. Engineering decisions, primary references and integrity appendix

STOP GATE: review before implementation.

No Stage-2/3/4 calculation, R1 production rule, saved revision or issued drawing is modified. Numerical completeness of an ideal-solid candidate is not engineering approval.

Five provenance classes: INHERITED; SYSTEM CALCULATED; SOURCE-BACKED REFERENCE; PRE-PILOT R1 ENGINEERING RULE; UNRESOLVED. In this successor report, a proposed engineering rule is *unapproved* unless explicitly described as an already adopted construction decision.

1 · Stage-3 audit — selection and candidate coverage

Software selection/integrity audit: PASS for the actual saved bounded search. The selected rotor/column ratio 0.33 is supported by the stored trials and exact offline replay; it is not an unevaluated default, first-iteration selection, or tie/weighting accident. This does **not** validate the underlying correlations outside their published ranges, prove a continuous/global optimum, or qualify the proposed shrouded/perforated successor hydraulically.

The evidence is an actual SELECT-only read of the application-connected database, not a UI fixture. Scientific record identity: design **269**, Stage-3 run **64**, Stage-4 calculation **13**. The two records have matching owner and design, without extracting owner PII. Run 64 is the latest Stage-3 run for this design at the time of the read. Run 63 has the same immutable hash; the duplicate is not a differing numerical result. This audit identifies the saved chain; it does not invoke an API that may create a current optimizer.

Important correction to the supplied working context: the actual saved speed is 45 RPM, not 70 RPM. Its selected feasible operating window is 30–60 RPM, not 60–105 RPM. Seventy RPM is the upper bound of the configured search grid. Both Stage 3 and the linked Stage 4 agree on 45 RPM. Do not use 70 RPM as the inherited operating point in the engineering report.

Saved engine: **ECR_STAGE3_STAGE4_OPTIMIZER_V1.3.0**.

Configured grids, all present in the stored full result:

- D: 0.2–1.5 m inclusive, 0.1-m steps (14).
- h_c/D : 0.20, 0.25, 0.30 (3).
- D_R/D : **0.33, 0.40, 0.50** (3).
- ϕ_s : 0.20, 0.30, 0.40 (3).
- RPM: 30–70 inclusive, steps of 5 (9).
- Two orientations, with the actual saved orientation retained as authoritative.

Each orientation has 378 geometry groups and **3402 stored trials**, total **6804**. Cartesian-key tests verified every configured combination exactly once, with no duplicates or missing combinations. Each rotor ratio has 126 groups / 1134 trials per orientation.

D _R /D	NMP-continuous feasible trials (all D/ h_c / ϕ /RPM)	Reverse-orientation feasible trials
0.33	596	341
0.40	318	74
0.50	102	0

These are actual persisted trial counts, not inferred from array lengths or just the declared candidate grid.

1.1 · Why the 0.33 rotor ratio won

The implemented policy is lexicographic, not a weighted objective:

- Determine feasible contiguous RPM windows for each geometry.
- Treat width ≥ 20 RPM as the useful-window **preference**, not an additional hydraulic equation.
- Choose the **second-smallest distinct accepted adequate diameter**, deliberately one accepted grid diameter above the smallest.
- At that diameter choose the widest window, followed only if needed by deterministic tie criteria.

The saved accepted adequate diameter set is 0.5, 0.6, ..., 1.5 m. Thus 0.5 m is the smallest adequate diameter and **0.6 m is deliberately selected**. This is an explicit pre-pilot margin policy, not an optimized economic diameter.

At $D=600$ mm in the inherited orientation:

Ratio	Best h_c/D , ϕ_s	Best feasible RPM window	Width	Meets 20-RPM preference
0.33	0.30, 0.40	30–60	30	Yes
0.40	0.30, 0.40	30–40	10	No
0.50	None feasible across all tested h_c/D and ϕ_s at $D=600$	None	—	No

At ratio 0.33 and $\phi_s=0.4$, $h_c/D=0.20$ and 0.25 both produce 25-RPM windows; $h_c/D=0.30$ produces 30 RPM. Therefore the selected complete geometry is a **strict width winner**, not a tie. The rotor ratio's late ascending tie-break clause is never needed for this winning decision.

The representative selected speed is the feasible trial nearest the **selected window midpoint**, hence $(30+60)/2 = 45$ RPM. The geometry-score power tie-break value is evaluated near the overall search midpoint (50 RPM); it is not the power at the selected 45-RPM point. Specifically, those values are 4.2146800763 and 3.0725017756 W/m³ respectively. Mixing the two would misstate the saved operating point.

At the selected 45 RPM, the stored candidate is FEASIBLE with no rejection reasons:

- Implemented actual loading = **0.47540999374941056**, below the design flood-fraction threshold **0.70**.
- Tip speed = 0.46652650905808435 m/s, below the implemented 4.5-m/s cap.
- $d_{32} = 0.0044853344930663765$ m; dispersed holdup = 0.06398240117180216.
- Power density = 3.072501775637703 W/m³; rotor Reynolds number = 21076.3029661.
- Positive geometry, power, d_{32} and interfacial area; holdup between 0 and 1.

At $D=600$ mm, $h_c/D=0.30$, $\phi_s=0.40$, the implemented loading gate explains the relative windows:

D_R/D	RPM	Actual loading	Result
0.33	60	0.6853015800	Pass
0.33	65	0.7616968758	Fail >0.70
0.40	40	0.6208041543	Pass

D_R/D	RPM	Actual loading	Result
0.40	45	0.7233596686	Fail >0.70
0.50	30	0.7015484950	Fail >0.70

This is model behavior, not evidence that smaller turbines universally perform better. The current primitive couples turbine power/droplet size, characteristic/swarm velocities and a maximized flooding-capacity construction. It contains fixed power/turbulence assumptions ($N_P=1.2$, $C=0.42$); the audit neither changes nor independently qualifies them for the successor turbine.

1.2 · Replay, order independence and applicability

- Canonicalization deduplicates/sorts and range-checks the configured ratio grid; stored controls explicitly include all three ratios.
- The source loops exhaust the Cartesian product; no first-feasible early-return bypass exists in this search.
- Reverse group order and 20 cyclic permutations produced the identical selected group in both orientations.
- Ranking uses ordered comparisons, **no weighted sum**. Window width strictly resolves this case before power or ratio tie-breaks.
- The final deterministic tie chain is edge margin (descending), valid trial count (descending), distance to grid-center speed (ascending), power density (ascending), D , h_c/D , D_R/D and ϕ_s (ascending). This can favor a smaller ratio only in otherwise tied cases; it does not explain the saved winner.
- Actual selected ratio is recomputed from the persisted controls and basis, not a display fallback. Stage-4 stored hydraulics match Stage 3.
- Full offline pure-function replay reproduced the **entire saved result**, including every trial and calculation hash, exactly after canonical key ordering. No API, service writer, optimizer persistence function or scientific code modification was used.

Proceed using inherited $D_R=198$ mm and 45 RPM for engineering development, but keep the hydraulic-qualification caveat explicit.

The selected trial is expressly SCALE_UP_EXTRAPOLATION, not a source-range-qualified design. Persisted diagnostics include:

- KH1995 source ranges exceeded for continuous density 1015 kg/m^3 , dispersed viscosity $0.0598 \text{ Pa}\cdot\text{s}$, $D=0.6 \text{ m}$, $D_R=0.198 \text{ m}$, $h_c=0.18 \text{ m}$, $h_c/D=0.30$ and $D_R/D=0.33$.
- In particular, the reported KH1995 D_R/D source range is 0.43–0.69, although the configured engineering search range includes 0.33–0.50.
- Myint terminal Reynolds 374.177, characteristic Reynolds 286.844 and shape Taylor 4.42967 extrapolation flags.
- A retained primitive diagnostic states that no generic Kühni flooding capacity is admitted; the optimizer separately applies its coupled resolver loading gate. This is not a license to present the latter as independently validated plant flooding capacity.

Thus the audit passes **algorithmic correctness against the implemented screening policy**, not universal Kühni performance, source applicability, or applicability of the inherited results to the changed shrouded/perforated geometry. The finite ratio grid cannot establish an interior optimum or justify extrapolating below its lower boundary. No new hydraulic equivalence claim is made.

2 · Inherited turbine OD and saved authority

Quantity	Actual saved value	Provenance
Column ID D	600 mm	INHERITED, Stage 3
Rotor ratio D_R/D	0.33	INHERITED, Stage 3
Overall turbine OD D_R	198 mm	INHERITED, Stage 3
Compartment pitch h_c	180 mm	INHERITED, Stage 3
h_c/D	0.30	INHERITED, Stage 3
Empirical stator free-area fraction ϕ_s	0.40	INHERITED, Stage 3
Selected speed	45 RPM	INHERITED, Stage 3
Selected contiguous feasible grid window	30–60 RPM , 5-RPM grid	INHERITED, Stage 3
Phase configuration	NMP continuous / RRBO dispersed	INHERITED, Stage 1/3
RRBO flow	4000 L/h (0.00111111111111111111 m ³ /s)	INHERITED, saved process basis
Wet-solvent flow	0.0005707717569786535 m ³ /s = 2054.778325 L/h	INHERITED; unit conversion SYSTEM CALCULATED
Temperature	40 °C	INHERITED
S/O mass ratio	0.6	INHERITED
Required / installed active height	7000 / 7020 mm	INHERITED, Stage 4
Compartment count	39	INHERITED, Stage 4
Fixed design N_t / screening HETS	7 / 1 m per theoretical stage	INHERITED, Stage 4

The discussion's 3500 L/h is not the saved RRBO flow; neither flow is a turbine circulation/pumping specification. The preliminary Ø44 shaft follows the existing Stage-5 geometric proportion $(10/45) \times 198$ and is not a Stage-3 shaft-strength selection.

Disposition: use $D_R = 198$ mm and 45 RPM for this engineering development. Do not reinterpret the audit as approval of the proposed physical construction.

3 · Complete turbine — construction and provenance

The adopted project topology is a double-entry, radial-flow, shrouded six-blade turbine. Primary source A (Asadollahzadeh et al., 2017, pp. 150–151, Fig. 1 and Table 1) directly supports the shrouded six-blade family. Double entry is the project construction decision. No inspected source supplies a numerical eye diameter.

Garthe Fig. 4.9 depicts exposed stepped blades and ring stators; its proportions are not a source for the proposed shrouds. Oliveira's 170 distributor holes are not stator holes. Full primary citations and caveats are in §11.

Provenance labels: **INHERITED** = Stage-3/4 input confirmed by the companion audit; **SYSTEM CALCULATED** = deterministic geometry from stated inputs; **PRE-PILOT R1 ENGINEERING RULE** = explicitly proposed, unapproved successor choice (not an existing approved software rule); **UNRESOLVED** = requires evidence or engineering approval. **SOURCE-BACKED REFERENCE** is deliberately not assigned to a dimension merely because similar historical equipment exists.

Quantity	Candidate	Provenance / rationale
Column inside diameter D	600	INHERITED, confirmed design269/run64
Overall turbine diameter DR	198	INHERITED, audited DR/D = 0.33
Compartment pitch hc	180	INHERITED, confirmed
Empirical stator fraction phi	0.40	INHERITED; mapping to physical gross area is an engineering convention, not hydraulic equivalence
RPM / feasible window	45 / 30–60	INHERITED from companion saved-result audit; not 70 RPM
Phase configuration	NMP continuous / RRBO dispersed	INHERITED from companion audit
RRBO net flow	4000 L/h	INHERITED; not equated to turbine circulation
Shaft ds	44	PRE-PILOT R1 ENGINEERING RULE: user-stipulated preliminary shaft, not strength-qualified or a new hydraulic selection
Turbine type	Double-entry radial-flow shrouded	Established user construction basis; no basic type re-selection
Blade count / spacing	6 / 60°	Established six-blade reference construction; geometrical spacing SYSTEM CALCULATED
Hub diameter / height	70 / 24	PRE-PILOT R1 ENGINEERING RULE: explicit review candidate; 13 mm radial hub material around the shaft, ample overlap with the proposed web. Attachment/key/fit/weld design UNRESOLVED
Blade thickness	3	PRE-PILOT R1 ENGINEERING RULE: trial plate gauge; stress, vibration, corrosion and joining not qualified

Quantity	Candidate	Provenance / rationale
Inner web axial height	16	PRE-PILOT R1 ENGINEERING RULE: reduced-height web to retain entry paths above and below while connecting the hub to outer paddles
Outer blade axial height	28	PRE-PILOT R1 ENGINEERING RULE: trial discharge height, independent of net process flow
Shroud thickness, each	2	PRE-PILOT R1 ENGINEERING RULE: geometric sheet envelope only; strength and distortion not qualified
Shroud radial width	34	PRE-PILOT R1 ENGINEERING RULE: explicit trial support/outer-paddle width; selection procedure below
Overall turbine height HR	32	SYSTEM CALCULATED: $28 + 2 \times 2$
Stator thickness	4	PRE-PILOT R1 ENGINEERING RULE: review plate envelope, not mechanical design; differs from old R1 $\min(D/160, hc/20) = 3.75$

The hub is not silently inherited from the former R1 1.6ds convention. The round 70 mm candidate is proposed here and must be approved independently; changing it changes all effective eye calculations.

3.1 · Ideal-solid topology and coordinate convention

Use a rotor-local Cartesian system: $Z = 0$ at the turbine midplane, Z positive upward, X/Y in the transverse plane, positive azimuth counterclockwise from $+X$ when viewed from $+Z$. Translate the rotor-local origin to each inherited rotor elevation only after approval. A typical compartment has stator centre planes at $Z = \pm 90$.

The candidate consists of:

- Continuous Ø44 shaft.
- Annular hub: radial interval $22 \leq r \leq 35$; axial interval $-12 \leq Z \leq +12$.
- Lower shroud: annulus $65 \leq r \leq 99$; $-16 \leq Z \leq -14$.
- Upper shroud: annulus $65 \leq r \leq 99$; $+14 \leq Z \leq +16$.
- Six 3-mm-thick radial blades at azimuths $0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ$.
- Each blade's inner web occupies $35 \leq r < 65$ and $|Z| \leq 8$.
- Each blade's outer paddle occupies $65 \leq r \leq 99$ and $|Z| \leq 14$.
- Both axial eye apertures are Ø130, bounded by the inner shroud edge. The upper and lower entries are geometrically identical.
- Discharge is through the six outer-periphery passages between the shrouds, not through drilled holes in the shrouds.

For an exact blade solid at azimuth α , rotate local coordinates (u,v) , with u outward along the blade and v transverse. The blade lies within $|v| \leq 1.5$, $u > 0$ and $35 \leq \sqrt{u^2+v^2} \leq 99$; its axial half-height is 8 for $r < 65$ and 14 for $r \geq 65$. Thus the root, step and outer edge are cylindrical-radius trimmed. There are no rectangular corners protruding beyond $DR = 198$. Joining surfaces at $r = 35$ and $Z = \pm 14$ are intentional interfaces, not collisions. Six blades are distinct; the opposite blade is not double-counted as part of one strip.

In the zero-thickness centre-plane sketch the positive-radius stepped blade outline is (r,Z) :

$(35,-8), (65,-8), (65,-14), (99,-14), (99,+14), (65,+14), (65,+8), (35,+8)$. Finite-thickness construction must use the radial clipping above, not extrude these coordinates into an oversized rectangular outer corner.

4 · Eye candidate — engineering choice, not a correlation

Neither the inherited variables nor gross free area uniquely determines DTI. The proposed construction defines it through a **chosen physical shroud/paddle radial width w**:

$DTI = DR - 2w.$

For review, consider $w = 30\text{--}38$: this provides an outer working/shroud span of 10–12.7 blade thicknesses and leaves an inner radial web reach of 26–34 from the Ø70 hub. These are declared geometric screening choices, not stress criteria, published Kühni proportions or an optimum. A nominal $w = 34$ balances those two spans and gives:

$DTI = 198 - 2 \times 34 = 130.$

The resulting DTI/DR ratio is merely an output (approximately 0.6566), not the sizing rule. Its unique acceptance remains UNRESOLVED pending engineering review and any primary-source dimensional evidence. Selecting a physical width does not eliminate the engineering assumption; it makes the assumption explicit and testable.

Trial shroud width	Calculated eye	Minimum eye-disk area in turning region, each side (diagnostic)	Conditional centre-opening candidate using \$4 rule
30	138	10,496.577	120
34	130	8,884.689	112
38	122	7,373.333	104

These are sensitivity alternatives, not three approved designs. Remaining calculations use the middle candidate only.

5 · Upper and lower entry sections

For DTI = 130 and ds = 44:

- $ATI_{gross} = \pi \times 130^2 / 4 = 13,273.229$.
- $A_{shaft} = \pi \times 44^2 / 4 = 1,520.531$.
- $ATI_{shaft-corrected} = 11,752.698$.

For this particular topology, the shaft-only number is the actual geometric effective area at each shroud eye face: the hub and blades do not reach those faces. It is not the same as every internal cross-section. The internal station analysis below is necessary but must not be mistaken for a series of identical-flow axial throats.

Inspect the eye disk $r \leq 65$ at each axial station. A solid boundary is evaluated using the fluid-side limit; coincident zero-thickness mathematical faces are not counted twice.

Upper station (lower is mirror image)	Actual obstruction union in eye disk	Open area
$Z > +16$; aperture at +16 through +14	Shaft only; shroud is outside $r = 65$	11,752.698
$+12 < Z < +14$	Shaft only	11,752.698
$+8 < Z < +12$	Shaft and hub jointly equal one Ø70 solid disk	9,424.778
$0 \leq Z < +8$	One Ø70 disk plus six non-overlapping blade strips outside the hub	8,884.689
$-8 < Z \leq 0$	Mirrored lower eye/web region	8,884.689
$-12 < Z < -8$	Hub/shaft union Ø70	9,424.778
-16 through -14; below -16	Shaft only in aperture	11,752.698

The hub begins at $Z = \pm 12$, inside the eye, and therefore changes the real entry area. The inner blade webs begin at $Z = \pm 8$. The shroud does not intrude into $r < 65$; there is no hypothetical full disk blocking entry. The taller outer paddle lies outside the eye disk.

5.1 · Obstruction union and unresolved limiting streamtube

For exact strip area, define:

$$F(R,t) = t \sqrt{R^2 - t^2/4} + 2R^2 \arcsin[t/(2R)].$$

F is the intersection area of a circle of radius R with the complete strip $|v| \leq t/2$, including both positive and negative u. Each actual radial blade occupies one half, so:

$$A_{\text{blade,eye}} = (6/2) [F(65,3) - F(35,3)] = \mathbf{540.089}.$$

$$A_{\text{eye-disk,turning-min}} = \pi(65^2 - 35^2) - A_{\text{blade,eye}} = \mathbf{8,884.689}.$$

There is no extra shaft subtraction after subtracting the Ø70 hub disk; the shaft is already contained in that disk. Similarly, only the blade strip outside the hub is subtracted. Blade roots do not overlap one another: 3 mm is far smaller than the 35 mm chord between adjacent blade centre lines at the hub radius.

Therefore:

Actual upper and lower effective eye-face area = 11,752.698 each, at Z = +16 and –16 respectively (and throughout each shroud aperture).

The downstream hub-cap cuts at Z = +12– and –12+ have **9,424.778** geometric open area each. They are actual internal approach/turning sections, not proven full-branch throats: the eye is not a closed axial duct and flow may already leave the eye cylinder radially above these cuts.

The conservative **minimum eye-disk geometric diagnostic = 8,884.689 each**, in the web/turning region $|Z| < 8$. It is explicitly **not** reported as a certified controlling entry throat. The minimum pre-turn, whole-branch streamtube section and corresponding hydraulic effective area are **UNRESOLVED**. Only a defined streamtube/control-volume analysis or resolved flow field can establish which internal cut intercepts all of each entry branch before it turns.

Important interpretation: these are minimum *geometric axial cross-sections of the entry/turning region*. The flow turns radially along that region, so the sections must not be modelled as an unbranched constant-throughput axial nozzle train. The station schedule identifies the actual solid obstruction, not a proven controlling hydraulic restriction or discharge coefficient. A single true flow-controlling throat cannot be certified without a flow solution/test. Equality of upper/lower geometric areas does not prove equal entry flow rates.

At r = 99, clear discharge height is 28. Subtract the six blade arc widths, not their projected eye areas:

$$A_{\text{radial,out}} = [2\pi \times 99 - 6 \times 2 \times 99 \arcsin(3/198)] \times 28 = \mathbf{16,912.970}.$$

At the eye outer edge r = 65+, where full-height outer paddles begin, the separate cylindrical turn-passage area is:

$$A_{\text{radial,turn}} = [2\pi \times 65 - 6 \times 2 \times 65 \arcsin(3/130)] \times 28 = \mathbf{10,931.353}.$$

This is the combined radial passage area over the full 28-mm height, not an axial eye area or a separate full-height area available to each side. A symmetric half-height partition would give half this number per side as a bookkeeping convention, but the two entry streams are not separated by a solid disc. A geometric area smaller than other sections is a warning for flow review, not proof of the controlling streamtube.

The outer cylindrical discharge area is smaller than twice the per-side minimum eye-disk diagnostic (17,769.378), and the turn area at r = 65 is smaller still. These different orientations, branching and flow turning prevent a simple smallest-area argument from certifying the actual hydraulic restriction. No pumping number, throughput equality, circulation multiplier or capacity guarantee is implied.

6 · Stator centre opening follows the eye

The adopted construction requirement is $DSO < DTI < DR$. The candidate adds a transparent, **unapproved and arbitrary comparative geometric screen**: the shaft-corrected stator centre area should be at most 95% of the smaller turning-region minimum eye-disk diagnostic. This is not a solved hydraulic design condition: that diagnostic is not a certified entry throat. The 5% allowance is a proposed comparison margin, not a hydraulic correlation, fabrication tolerance or proof of capture.

$$DSO_{\max} = \sqrt{ds^2 + 4 \times 0.95 \times A_{\text{eye-disk, turning-min}} / \pi}$$

$$= 112.617585.$$

For a simple preliminary centre-opening callout, round downward to an even millimetre:

$$DSO = 112; CE = (130 - 112) / 2 = 9.$$

This rule is explicitly additional engineering judgment, not a claim that the documents or literature require a 0.95 factor. It is not a universal DSO/DTI ratio. Engineering review may reject it; its sensitivity is visible.

- $ASO_{\text{gross}} = \pi \times 112^2 / 4 = 9,852.035.$
- $ASO_{\text{shaft-corrected}} = \pi(112^2 - 44^2) / 4 = 8,331.504.$
- $ASO_{\text{shaft-corrected}} / A_{\text{eye-disk, turning-min}} = 0.937737.$
- Actual comparative area allowance = **6.2263%** after rounding.
- Shaft-to-centre-opening radial clearance = $(112 - 44) / 2 = 34.$

The centre opening lies inside the eye's gross envelope in axial projection, but it is not a duct or sealed coupling. The stator and turbine are separated axially. A positive CE does not establish streamtube capture or control which perforation flows reach the turbine.

7 · Gross-area target and perforation development

$$AC = \pi \times 600^2 / 4 = \mathbf{282,743.339}.$$

Inherited $\phi = 0.40$ is mapped under the stated physical gross convention:

$$A_{\text{target}} = \phi AC = \mathbf{113,097.336}.$$

Required hole area after the eye and centre selection:

$$A_{\text{holes}} = A_{\text{target}} - A_{\text{SO,gross}} = \mathbf{103,245.301}.$$

Use four rings with counts 12, 18, 24 and 30, totalling 84 holes. Each ring count is a multiple of six. The rings and their phase shifts reserve six continuous radial no-hole lanes. The choices are explicitly proposed layout decisions, not a reused 54×Ø40 or two-quarter-point pattern.

Proposed preliminary layout goals: hole-to-hole ligament at least 12; centre edge ligament at least 20; shell edge ligament at least 15; six 8-wide no-hole radial lanes with at least 2 separation to the nearest hole. These are geometric screening goals only, not allowable structural values. Selected centres $r = 100, 155, 210, 265$ give uniform 55 radial pitch and pass these goals.

With $N = 84$:

$$d_h = \sqrt{[(0.40 \times 600^2 - 112^2) / 84]} = \mathbf{39.5594790278}.$$

The nonstandard diameter is a profile-cut CAD diameter, not an assertion that a standard drill exists. A released drawing needs fabrication method, tolerances, deburring and an acceptance rule for gross-area deviation. Ø39.56 is a display rounding only; use the unrounded calculation for exact nominal closure. If a standard drill diameter is mandated, this candidate must be rebalanced and re-reviewed.

Row	PCD	Count	First hole angle	Increment	Chord pitch	Arc pitch	Circumferential ligament
1	200	12	15°	30°	51.763809	52.359878	12.204330
2	310	18	10°	20°	53.830935	54.105207	14.271456
3	420	24	7.5°	15°	54.821001	54.977871	15.261522
4	530	30	6°	12°	55.400086	55.501470	15.840607

For row i , hole $j = 0 \dots N_i - 1$:

$$\theta_{ij} = (j + 0.5) 360^\circ / N_i;$$

$$X_{ij} = (PCD_i / 2) \cos(\theta_{ij});$$

$$Y_{ij} = (PCD_i / 2) \sin(\theta_{ij}).$$

This is the full pattern definition, not a representative subset. All 84 centres are provided in geometry-proposal.json. Row staggering is the explicit first-angle schedule above, not an undocumented common angle.

7.1 · Gross/net closure and support obstruction

$$DSO^2 + N d_h^2 = 112^2 + 84 \times 39.5594790278^2 = \mathbf{144,000}.$$

- Physical gross free area = **113,097.336**.
- Physical gross fraction = **0.400000**.
- Shaft-only adjusted stator free area = **111,576.805**.
- Shaft-only adjusted fraction = **0.3946222222**.

This candidate places no separate support bar across the centre opening or a hole at the stator plane. The six radial lanes are retained solid plate, already outside the counted openings; subtracting them again would double-count blockage. Wall attachment and final support details are unresolved. Any later support masking a hole or centre opening must have its *intersection with the already-open area* deducted as an additional diagnostic and be checked at its actual Z station. Separate shaft-support planes must not be subtracted from the stator plane as though they were coincident.

The old Ø379.473 central opening is not used. The new centre opening supplies only part of the target; holes supply the remainder. No shaft-adjusted value is returned to Stage 3.

Retired for this successor: Ø379.473 mm as a centre opening. It alone consumes the full 40% gross area. The historical single-opening R1 dataset remains unchanged.

8 · Clearance and ligament validation

The accompanying standalone calculation enumerates all 3,486 hole pairs, not just pairs in the same row. It calculates centre distances and subtracts the common hole diameter. Each row is also checked against the centre boundary, shell and six radial lane boundaries.

Check	Result	Status / limitation
DSO < DTI < DR	112 < 130 < 198	Geometric pass
Positive capture allowance CE	9	Geometric pass; not hydraulic capture proof
Hub-eye radial interval	30	Positive entry annulus
Rotor-wall radial clearance	201	Geometric pass
Stator planes relative to rotor	±90	From $h_c/2$
Stator inner faces	±88	With proposed 4 plate
Rotor extremes	±16	With proposed 32 HR
Rotor-to-stator axial clearance, each side	72	Geometric pass
Web top to shroud inner-face plane	6	Step space; radial positions differ at the boundary
Hub cap to shroud inner-face plane	2	Axial plane offset only; not a collision clearance between overlapping radial solids
Minimum same-row and global hole-pair ligament	12.204330	Row 1 adjacent holes
Minimum cross-row centre distance	55.345639	Between rows 3 and 4
Minimum cross-row ligament	15.786160	All cross-row pairs checked
Centre-opening to nearest hole ligament	24.220260	$r_1 - d_h/2 - DSO/2$
Outer hole to shell ligament	15.220260	$300 - r_4 - d_h/2$
Six reserved radial lane widths	8	Solid geometric lanes, not structurally qualified spokes
Minimum hole edge to lane boundary	2.102165	Row 1; exact $r \sin(\pi/N) - d_h/2 - 4$
Sixfold plate rotational symmetry	exact	Counts and phases repeat at 60°
Total gross area closure	0.400000	Nominal arithmetic pass
Rotor swept diameter	198 exact	Cylindrical clipping includes finite blade thickness
Upper/lower eye obstruction topology	mirrored	Equal geometry, not demonstrated equal flux

The 72-mm axial gap uses stator **centre-plane** pitch, consistent with the existing R1 stack definitions inspected read-only in shared/ecr-stage5-r1.ts: stators $z_0 + i h_c$, rotor centres $z_0 + (i + 0.5) h_c$ (manifest lines 18–21 and compartment mapping lines 211–212). This is a verified existing geometry convention, not a new inference from the database.

No-hole lanes are strips about radial axes at $0^\circ, 60^\circ, \dots, 300^\circ$, from the centre-opening edge to the outer plate. Because every ring has an even number of holes per 120° and half-step phase, the nearest hole on each side of a lane has distance $r \sin(\pi/N)$ to the lane axis; the table verifies actual lane clearance. These are not additional pieces intruding into the opening.

8.1 · Conditional quantities and assembly interfaces

The confirmed Stage-4 count is **39 compartments**, installed active height $7020 = 39 \times 180$. If the existing one-turbine-per-compartment and **N+1 shared stator boundaries including two terminal stators** convention is retained, the proposed complete active stack contains:

Component	Conditional stack quantity	Basis
Turbine assemblies / hubs	39 / 39	One per inherited compartment
Upper shrouds / lower shrouds	39 / 39	Two per turbine, 78 total
Individual radial blades	234	39×6
Stator plates	40	N+1 retained terminal-boundary convention
Stator perforations	3360	40×84 ; centre openings excluded
Continuous main shaft	1	Existing common-shaft architecture; length/support verification separate

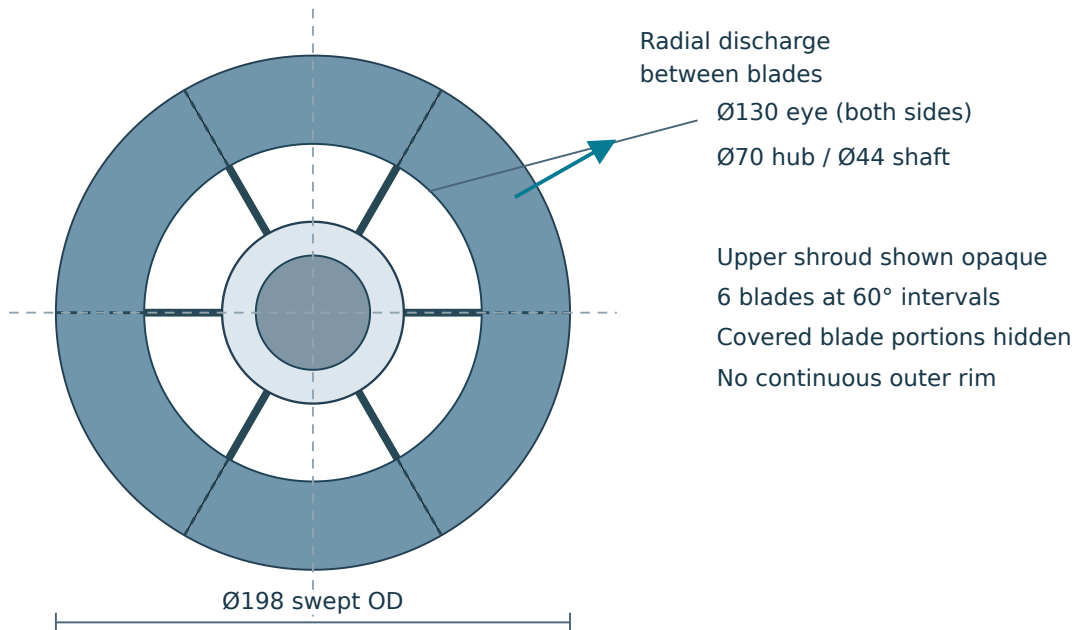
The N+1 count is a retained existing R1 engineering convention, not a universal published Kühni count. Terminal stators may require separate treatment; approval of the repeating pattern does not automatically approve terminal support/flow conditions.

If the existing R1 $z_0 = 1.25D = 750$ stack datum is retained, proposed rotor centres are $z_i = 750 + (i+0.5)180$, $i=0 \dots 38$, i.e. 840 through 7680; stator centres are $z_j = 750 + j180$, $j=0 \dots 39$, i.e. 750 through 7770. These formulas give all component elevations without reselecting counts or active height.

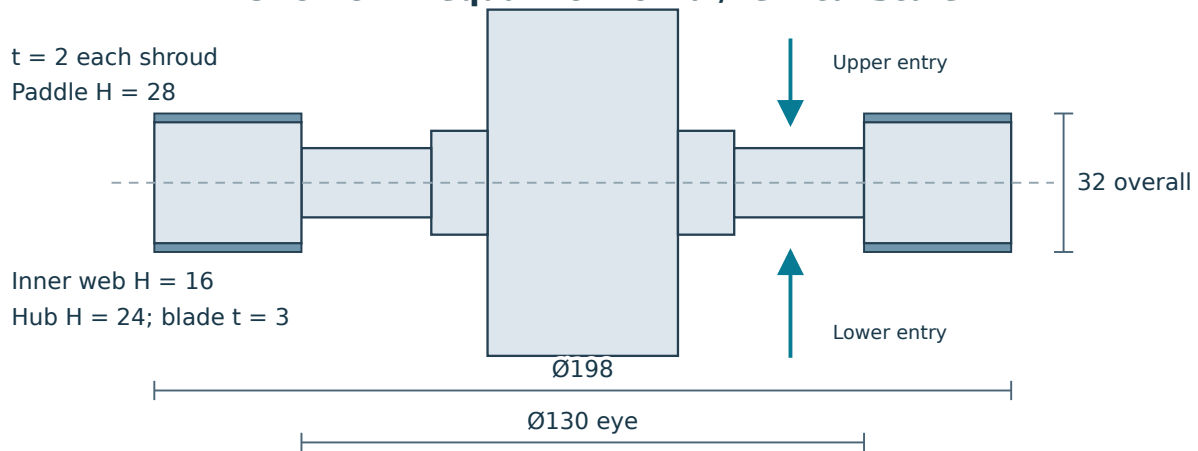
No saved Stage-5 snapshot/support geometry was supplied in the audit evidence available for this appendix. Therefore **complete-assembly collision approval is not claimed**. The read-only current R1 source has end-zone support centres at $0.75D$ and $z_1 + D/2$, with axial half-depth $0.02D$. Conditionally retaining that source convention gives centres 450 and 8070, axial envelopes [438,462] and [8058,8082]. With proposed 4-mm terminal stators, their nearest faces are 748 and 7772, leaving 286 axial separation to the respective supports; first/last rotor envelopes also remain separated. This is a source-convention compatibility calculation, **not verification against an actual immutable Stage-5 snapshot**. Final saved support housings, arms, nozzles, attachments and interface coordinates must be compared with the successor solids before a complete assembly can be approved. Do not subtract these separate end-zone support projections from active-stator plane areas.

9 · Proposed turbine drawing

T-01 · Proposed turbine — plan and blade-plane section



BLADE-PLANE SECTION • equal horizontal/vertical scale



Solid blade section: radial flow occurs out of this section between blades.

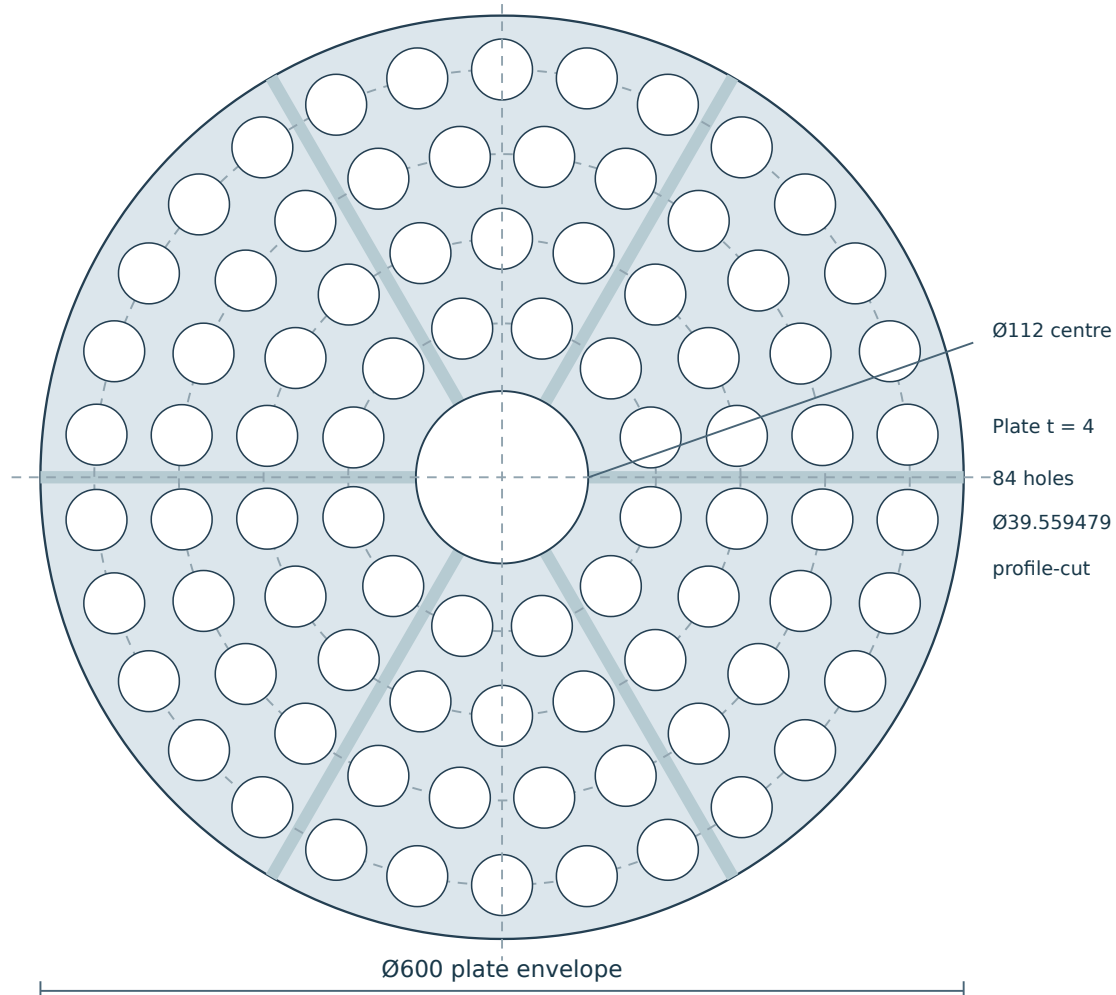
All new construction dimensions are proposed engineering choices; swept OD is inherited. Blue shrouds attach to the outer paddles at $Z = \pm 14$. Blade $t = 3$ is trimmed at cylindrical radii 35, 65 and 99. The section shows solids through two opposing blades; radial passage flow is between blades, as indicated in the plan.

9.1 · Proposed stator drawing

S-01 · Proposed perforated stator — complete 84-hole pattern

+X at right; positive azimuth counterclockwise viewed from +Z

Six 8-mm radial no-hole lanes: 0°, 60°, 120°, 180°, 240°, 300°



PCD: 200 / 310 / 420 / 530; counts: 12 / 18 / 24 / 30

First angle: 15° / 10° / 7.5° / 6°; radial centre pitch = 55

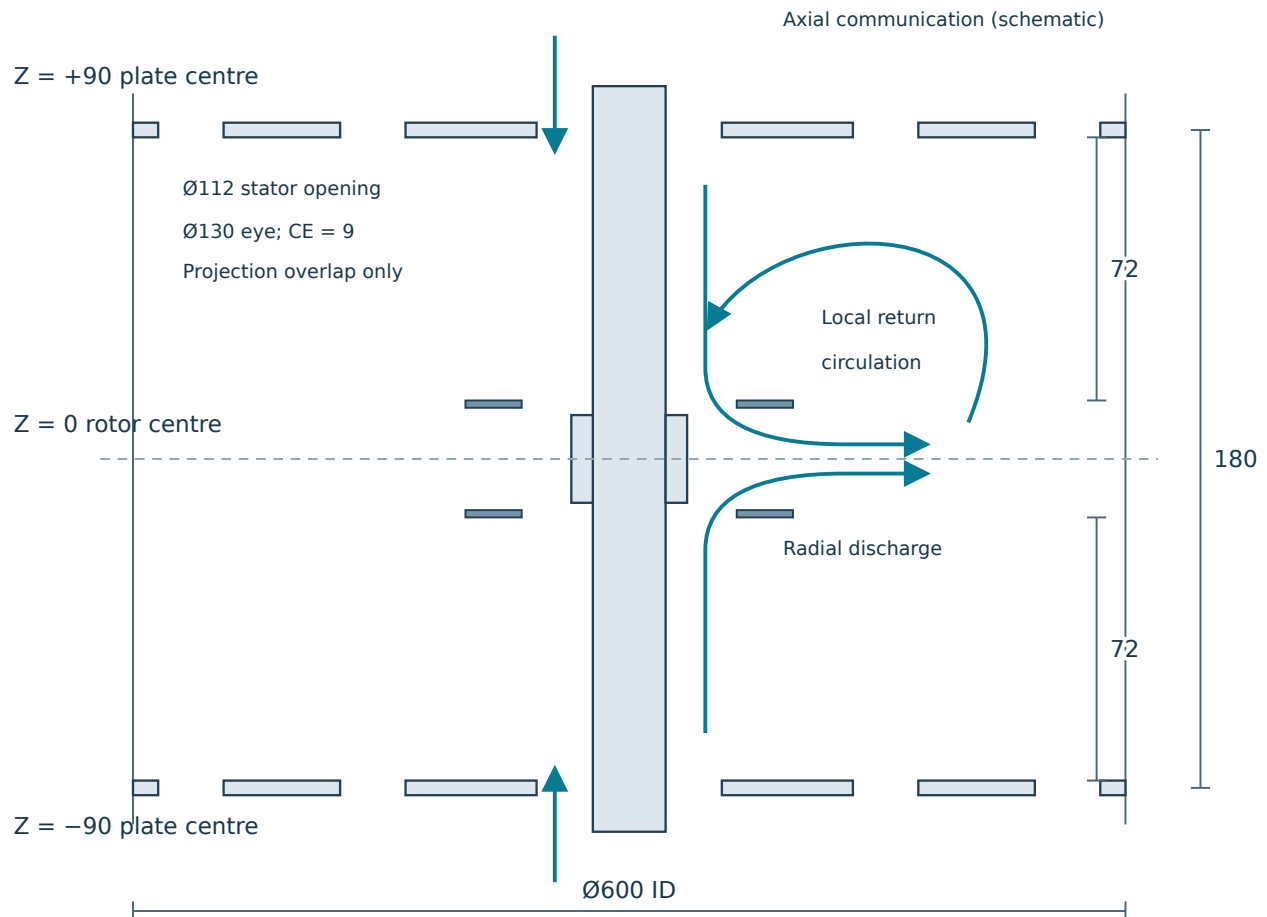
Shaded lanes are retained plate, not separate support bars; no extra area deduction.

Hole count and row layout: proposed engineering rules. Hole diameter: SYSTEM CALCULATED from exact gross closure. Nominal hole diameter shown rounded; use 39.5594790278 in review calculations. Ligament acceptance and plate strength remain unqualified. The no-hole lanes are not additional pieces.

9.2 · Proposed typical-compartment section

C-01 · Typical compartment — section between blade planes

Horizontal and vertical scales differ. Dimensions govern; do not scale.



Rotor outer faces $Z = \pm 16$; stator inner faces $Z = \pm 88$; plate $t = 4$

30° section cuts row-2 and row-4 holes; row-1/row-3 holes are out of section.

Section at 30° between blade planes intersects row-2 and row-4 perforations through their centres; other blades/perforations away from this cut are intentionally omitted. Flow arrows show the adopted architecture, not solved streamlines or imposed equal upper/lower flow. Stator opening and eye are not a sealed duct. Wall attachment and bypass sealing remain unresolved.

10 · Engineering decisions and release stop gate

- Acceptance of the specific 34-mm shroud width / Ø130 eye remains **UNRESOLVED**: the numerical candidate is a deliberately exposed engineering choice, not a sourced unique solution.
- Hub dimensions, blade plate gauge and stepped heights, shroud gauge, 5% comparative entry-area allowance, even-mm centre rounding, stator thickness, ring placement, lane/ligament screening criteria and fabrication method all require approval.
- The shape and dimensions are geometrically connected and collision-free for the stated ideal solids. They are not qualified for rotational strength, weld fatigue, balance, critical speed, erosion/corrosion, thermal expansion, deflection, pressure loading or shaft attachment.
- Weld beads, fillets, hub fasteners, surface finish, runout, tolerances and real shroud transitions are absent. Their real intersections may reduce effective areas; recalculate after they are defined.
- Station-area arithmetic is not a hydraulic performance validation. Local loss coefficients, nonuniform inflow, radial passage pressure gradients, operating power/pumping characteristics, phase holdup and backmixing remain unvalidated.
- A separate radial discharge area does not by itself identify the actual hydraulic minimum passage. Do not impose $Q_{\text{turbine}} = \text{process throughput}$ or infer pumping capacity from any of these area values.
- Equal gross phi does not demonstrate hydraulic equivalence between the old annular stator and this perforated design. The empirical Stage-3 model is unchanged, but the successor physical mapping needs independent qualification.
- All assembly supports, end-compartment differences and any inherited shaft-support elevations must be reconciled by the full report before a new complete rule set can be issued.
- The companion audit confirms the inherited dimensional tuple, 45 RPM and NMP-continuous/RRBO-dispersed configuration. That software audit does not approve these new engineering choices or qualify the successor hydrodynamics.

Stop after engineering review. No production calculations, persisted records, R1 datasets or drawings have been changed by this standalone calculation.

Reproduction: `node deliverables/kuhni-successor-review/geometry-proposal.mjs`. The script imports only Node filesystem, writes its own report JSON, asserts area closure and positive clearances, and does not invoke application modules, databases or optimizer services.

Review outcome required: approve, amend or reject the dimensional candidate and its explicit geometric assumptions. A successor rule-set version may be assigned only after review. No automatic production implementation is authorized by this report.

11 · Primary construction references

G — Dirk Garthe, Fluidynamics and Mass Transfer of Single Particles and Swarms of Particles in Extraction Columns, TU München dissertation

- Primary local file: attached_assets/Paper_1788587264307.pdf.
- Repository record: <https://mediatum.ub.tum.de/601973> (cached project HTML reports a bot-protection page; the local supplied primary PDF, not that HTML, was inspected).
- SHA-256: 39a39035d63f0b84bc74957ccf17b7a8c67ec456e73e4277491bc2cbb2bd53fc.
- Printed pp. 63–64, §4.5 and Figure 4.9 (PDF pages 75–76): the Kühni compartment illustration is bottom-right. It depicts annular stators, a common shaft and six stepped-looking radial blades. The illustrated blades are exposed; a pair of annular rotor shrouds is **not visibly established by this figure**. This is a limit on using this particular illustration as a shrouded-turbine manufacturing reference, not proof that shrouded Kühni turbines do not exist.
- Printed p. 177, Table A.1 (PDF p. 189): Kühni six-blade agitator diameter 45 mm, height 7 mm, relative stator free area 0.40, compartment height 50 mm, shaft diameter 10 mm.
- No turbine-eye diameter, rotor-shroud ID/OD/thickness, blade thickness, hub OD/height, stepped-web numerical proportions, or turbine-to-stator clearance dimension is supplied in that table.
- The table's 1- and 2-mm sieve-tray thicknesses and 2- and 4-mm perforation diameters belong to **different pulsed sieve-tray internals**. They must not be silently assigned to the Kühni stator.
- The 7/45 height ratio and 10/45 shaft ratio are calculated apparatus ratios, not universal scale laws. Applying them at OD 198 gives 30.8-mm height and 44-mm shaft, but those extrapolations are **PRE-PILOT ENGINEERING RULES**, not source-backed target dimensions. Adding shrouds also requires defining whether “height” includes them; Garthe does not settle that for the proposed successor.

11.1 · Oliveira and shrouded-family evidence

O — N. S. Oliveira, D. Moraes Silva, M. P. C. Gondim, M. Borges Mansur (2008), A Study of the Drop Size Distributions and Hold-Up in Short Kühni Columns

- *Brazilian Journal of Chemical Engineering*, 25(4), 729–741.
- Primary local file: attached_assets/A_STUDY_OF_THE_DROP_SIZE_DISTRIBUTIONS_1788585700948.pdf.
- SHA-256: 5381a0eb1d1ea54d6f3eacabf644364306d4bcbd0edf226191d6d4947be39317.
- Printed p. 731, “Experimental / Kühni Pilot Scale Unit” (PDF p. 3): five stages, column diameter 150 mm; six-blade rotor diameter 85 mm and height 10 mm; stator fractional free area 30%; stator spacing 70 mm. Internal parts AISI 316; annular Teflon gaskets prevent bypass between stator and wall.
- Apparatus ratios: $85/150 = 0.5667$; $10/85 = 0.11765$; $70/150 = 0.4667$. These differ from Garthe and do not select target DTI, HR, hub or shroud dimensions.
- The same paragraph describes **170 holes of 1-mm diameter on five concentric circles in the inlet distributor**, not holes in each stator. Do not repurpose the distributor pattern as the proposed stator pattern.
- The inspected apparatus text supplies no eye diameter or detailed shroud/hub/blade-entry dimensions. “Six-blade rotor” alone is not a dimensional shroud specification.
- Its water-continuous/Exxsol-dispersed system must not replace the saved project's phase configuration.

A — Asadollahzadeh et al. (2017), “Experimental Determination of Continuous Phase Overall Mass Transfer Coefficients: Case Study: Kühni Extraction Column”

- *Iranian Journal of Chemistry and Chemical Engineering*, 36(5), primary article.
- Primary local file: research/sources/kuhni-mass-transfer/asadollahzadeh-2017-primary.pdf.
- Primary source URL recorded with the project copy: https://ijcce.ac.ir/jufile?ar_sfile=344118
- SHA-256: 317c207d4fa0618f123b14239794aab51c1f29ddba217b91a08af2e532190239.
- Printed p. 150 (PDF p. 2), “Description of the Equipment”: explicit **50-mm diameter shrouded six-blade turbine agitator**, ten stages. This is direct primary support for a shrouded six-blade Kühni construction family.
- Printed p. 151 (PDF p. 3), Table 1: shrouded six-bladed turbine diameter 0.05 m; working height 0.7 m; stator fractional free area 0.212; ten compartments.
- Figure 1 on p. 151 includes an enlarged compartment sketch with perforated stators and a visibly shrouded rotor. It is a **schematic**, not a dimensioned component drawing: do not measure or scale its eye, hub, blade geometry, stator hole count or PCD.
- Important source inconsistency: prose says column internal diameter **113 mm**, while Table 1 says column diameter **0.117 m**. Preserve this discrepancy; neither is needed to infer the target eye.
- The inspected paper supports the construction family and a perforated-stator illustration, not a universal DSO/DTI relation, total free area of 0.40 for the successor, or published 198-mm turbine dimensions.

11.2 · Source limits and classification

There is primary support for shrouded six-blade Kühni turbines, and a primary illustration of perforated stators. The accepted user topology therefore need not be reopened. However, **Garthe's illustrated exposed-blade geometry cannot be presented as a dimensioned source for the new upper/lower shrouds.**

No inspected primary source establishes a numerical turbine eye DTI, eye/OD ratio, upper/lower eye thickness, lip radius, entry-blade start radius, or capture margin CE for this proposed 198-mm turbine. A limited additional web search returned no usable dimensioned primary eye source; unrelated pump/compressor/wind-turbine sources were rejected. This is an evidence gap, not a claim that such evidence cannot exist.

Consequently:

Item	Valid report classification
D, DR, hc, RPM, empirical phi and phase orientation from verified saved Stage 3	INHERITED
Six blades and shrouded construction family	SOURCE-BACKED REFERENCE, also adopted project topology
Explicit double entry and turbine-first successor sequence	Adopted PRE-PILOT ENGINEERING RULE / user design basis
Garthe 45-mm rotor, 7-mm height, 10-mm shaft in its own apparatus	SOURCE-BACKED REFERENCE
Scaling those to HR = 30.8 mm and ds = 44 mm for DR = 198 mm	PRE-PILOT ENGINEERING RULE (shaft not strength verified)
New hub, shroud thickness, stepped inner web, eye and DSO values	Proposed PRE-PILOT ENGINEERING RULE with rationale, or UNRESOLVED
DSO < DTI < DR and positive CE	Adopted PRE-PILOT ENGINEERING FLOW-PATH RULE, not literature correlation
Area and ligament results from fully defined proposed solids	SYSTEM CALCULATED
Pumping number, circulation ratio, hydraulic equivalence, shaft adequacy	UNRESOLVED / outside demonstrated evidence

Sources were visually inspected at the cited figures/tables. Their numerical dimensions refer to their own apparatus. The new 130-mm eye, 112-mm centre opening and 84-hole layout are not claimed as published Kühni or proprietary Sulzer dimensions.

12 · Scientific integrity and reproducibility

- Stage-3 immutable hash, independently reconstructed and verified:
e2afc059444f3c22c5d3a64b83ed81b59b5a122bd51ac1c9bc9897337d788249.
- Stage-3 engine/implementation hash:
faa56fdd99b0904a5e524fb8350859192d94b269331e91bbd34cc5d946ccb550.
- Stage-3 calculation hash, saved = replay:
f7cacea88792a9b70aecb65e187728c5e4aae1fa038920a450d9e2385bf850ef.
- Full result canonical SHA-256, saved = replay:
7dc0cc2a2350bb06b4c024ac82bd86a21d7efacb57164094695dd8c4447a9ebf.
- Stage-1 snapshot hash: 549dffa428356ee5e94d8baff297c114e1f0c6f345b84e121181e7b5f6b5f15b.
- Stage-4 ID 13 lineage hash: 5a9a6446a4e59faff9d5a86ee02f21afbaf11b1524e02641d5dd449424d380b3.
- Stage-4's stored Stage-3 immutable hash equals the verified run-64 hash.
- SELECT export file SHA-256 before and after the final offline replay/re-read:
a20f1b5039e3b9f56c99dd8964f126e39c5ee5e9b4ce184910e947785f9e7fc8 both times.

The export includes Stage-3/4 scientific records and comparison IDs, not credentials or owner personal data. Different hashes above represent different envelopes and must not be substituted for one another. Stage-4's optimizerResultHash is its projected/compacted representation identity, not the full raw Stage-3 calculation hash.

Evidence files in this directory:

- saved-evidence.json: actual saved records and owner/design-match booleans.
- ranking-evidence.json: exact replay identities, Cartesian completeness, ratio counts, ordering tests and all D=600-mm group summaries.
- audit-read.ts: direct database SELECTs inside BEGIN READ ONLY, rolled back after reading.
- audit-replay.ts: imports only pure scientific functions; no database access or writes except review evidence files.

12.1 · Audit code citations and calculation artifacts

Source citations:

- server/ecr-pre-pilot/stage3-stage4-optimizer.ts:76-86: configured grid and fixed 20-RPM preference.
- Same file :340-378, :381-466: deterministic canonicalization and bounded grids.
- Same file :536-596, :665-735: candidate physical checks and hydraulic loading admission.
- Same file :775-784, :870-880, :950-977, :1043-1085: tie order, width ranking, second-smallest adequate diameter.
- Same file :1091-1117: selected RPM midpoint policy.
- Same file :1120-1206: exhaustive loops, contiguous windows and geometry score construction.
- Same file :1358-1384: authoritative orientation retained; alternate orientation is comparison, not an automatic phase switch.
- Same file :1525-1531, :1572-1589: pure optimizer entry and API-only compaction (stored trials retained).
- server/ecr-pre-pilot/kuhni-geometry-resolver.ts:180-242: d32, characteristic/swarm velocity and implemented loading construction.
- server/ecr-pre-pilot-service.ts:491-547: authority loading, full-result immutable envelope and persisted-result/API-compaction separation.
- server/ecr-pre-pilot/stage4-pre-pilot-sizing-service.ts:605-658: current saved record lookup and immutable/calculation integrity checks.

Stop gate: this audit authorizes **no implementation change**. Detailed eye, shroud, hub, stator and perforation dimensions remain the engineering report's next work, with provenance and unresolved items stated explicitly.

Companion reports: stage3-audit.md; source-review.md; geometry-proposal.md. Standalone geometry reproduction: node deliverables/kuhni-successor-review/geometry-proposal.mjs. Complete 84-hole coordinates and derived quantities: geometry-proposal.json. Renderer: build-report.mjs. These are report-only artifacts, separate from production models.